**Week 12:** 16 – 20 November 2026

**Subject:** Mathematics **Unit Title:** Statistics **Grade :** 12DP

| Lessons الحصص | Learning targets أهداف التعلم | Assessment tools and strategies التقييم Formative تكويني/ Summative ختامي / Differentiation التمايز | Resources and links for Parents و المصادر الروابط المخصصة لأولياء الأمور |
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| **Lesson 1 Scatter Diagram, Linear Regression & Line of Best Fit** | • Construct a **scatter diagram** for bivariate data.<br>• Use technology to determine the **line of best fit (y = a + bx)**.<br>• Interpret the **slope and intercept** in context.<br>• Predict outcomes using regression equations. | 1.1 Flexible display of information (size, color, font, layout).<br>2.1 Clarify vocabulary and symbols using visuals, connections, and supports.<br>8.1 Make goals clear, visible, and supported by short-term objectives. | Reflection prompts, notebooks<br>The resources will be available on student's Teams channel<br><br>Revision Village problems based on variations (direct and indirect) and geometry of 3D shapes.<br>https://www.revisionvillage.com/<br>Past IB AI SL questions on Statistics.<br>https://www.savemyexams.com/members/<br>InThinking online assessment to measure student progress on the concept understanding of Statistics.<br>https://thinkib.net/mathapplications |
| **Lesson 2 Tree Diagrams and Probability** | • I can **analyze** probability situations by **constructing** tree diagrams and using them to **determine** the probabilities of events. | | |
| **Lesson 3 Venn Diagrams and Overlapping Events** | • I can **analyze** overlapping events by using Venn diagrams to **represent** relationships between events and **calculate** probabilities. | | |
| **Lesson 4 Venn Diagrams and Overlapping Events** | • I can **analyze** overlapping events by using Venn diagrams to **represent** relationships between events and **calculate** probabilities.<br>• | | |
| **Teacher's notes/ Reminders ملاحظات المعلم والتذكير** | | | |

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| **Technology Integration دمج التكنولوجيا** | Use Desmos, GeoGebra and Excel for data visualization, boxplots and frequency tables; Teams for sharing and reflection. |
| **Cross-curricular links الروابط بين المواد الدراسية** | Science (data analysis of experiments), Economics (market sampling), Geography (population data comparison). |

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| **External Assessment Integration** دمج التقييمات الخارجية | All content is covered in IB Past Exams |
| **Wellbeing/ ATL Skills** الصحة النفسية/ أساليب التعلم | **Wellbeing:**<br>• Confidence in solving real-world maths problems.<br>• Resilience when learning from mistakes.<br>• Collaboration through shared investigations and discussions.<br>• Self-awareness by setting and reflecting on learning goals.<br>**IB ATL Skills:**<br>• **Thinking Skills** – Interpret numerical data to identify patterns and trends.<br>• **Communication Skills** – Present and explain statistical findings clearly.<br>• **Self-Management Skills** – Plan and execute data collection and organization.<br>• **Research Skills** – Use authentic data sources to analyze results.<br>• **Collaboration Skills** – Work in groups to design surveys and interpret results. |
| **Standards** المعايير | IB Math AI SL Standards:<br>• 7.SP.A.1-2<br>• 7.SP.B.3<br>• HSS-ID.A.1-2<br>• 8.EE.B.5-6<br>• 8.F.B.4<br>• HSA-CED.A.2<br>• HSF-BF.A.1 |
| **Inquiry/ TOK Question** البحث والاستقصاء أسئلة **TOK** | How can we trust data to tell the truth if the method of collecting it is biased? |
| **Success Criteria** معايير النجاح **(Measurable)** | Lesson 1:<br>▨ **All Students Must:** construct and interpret a scatter diagram for bivariate data.<br>▩ **Most Students Should:** use technology to find and interpret a line of best fit, including its slope and intercept.<br>▲ **Some Students Could:** use a regression equation to make predictions and assess the reasonableness of those predictions in context.<br>solve questions involving algebra, functions, geometry & trigonometry, and statistics & probability.<br><br>Lesson 2: |

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| | **All Students Must:** Construct a tree diagram that shows all possible outcomes of an event.<br>**Most Students Should:** Use the tree diagram to calculate and explain probabilities of simple and compound events.<br>**Some Students Could:** Compare different probability scenarios, justify their reasoning, and evaluate the efficiency of using tree diagrams to solve probability problems.<br><br>Lesson 3:<br>**All Students Must:** Identify and represent events and their overlaps accurately in a Venn diagram.<br>**Most Students Should:** Calculate and explain probabilities using information shown in a Venn diagram.<br>**Some Students Could:** Analyze complex probability situations involving overlapping events and justify conclusions using evidence from the Venn diagram.<br><br>Lesson 4:<br>**All Students Must:** Identify and represent events and their overlaps accurately in a Venn diagram.<br>**Most Students Should:** Calculate and explain probabilities using information shown in a Venn diagram.<br>**Some Students Could:** Analyze complex probability situations involving overlapping events and justify conclusions using evidence from the Venn diagram. |
| **National Identity Domain**<br>**مجالات الربط بالهوية الوطنية** | Belonging الإنتماء Respect الاحترام Heritage التراث |